

A cyclic-partition bijection for improper partition matrices

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1 The original question

Chern and Fu's Question 5.7 [1] asks for a direct bijection between two classes, preserving semi-weight on one side and the number of distinct values on the other. Here is the complete problem, including its boundary conventions.

A *partition matrix* on $[n] = \{1, \dots, n\}$ is an upper-triangular square matrix of sets. Its nonempty cells partition $[n]$, every row and column is nonempty, and column indices are nondecreasing as the labels $1, \dots, n$ are read. Thus the labels of each column form an interval. An ascent or descent is a pair of consecutive labels $i, i+1$ in one column but different rows. It is *proper* when i has the same parity as the smallest label in that column. A matrix is *improper* if it has no proper ascent or descent. Write IPPM_n for these matrices and put

$$v(Q) = \sum_{j=1}^{\dim Q} \left\lceil \frac{N_j}{2} \right\rceil,$$

where N_j is the number of labels in column j . The subclass IPPM_n^- consists of matrices whose cell containing n has odd cardinality.

An inversion sequence is $e = (e_1, \dots, e_n)$ with $0 \leq e_i < i$. Let $\mathbf{I}_n(-, -, =)$ consist of those with no $i < j < \ell$ such that $e_i = e_\ell$. Equivalently, every value occurs in one consecutive run of length one or two. Let $\text{dist}(e)$ count all distinct values, including zero. The subclass $\mathbf{I}_n^-(-, -, =)$ consists of sequences whose last run has length one. The unique objects at $n = 1$ belong to both minus classes.

Original problem. For every $n \geq 1$, construct mutually inverse maps

$$\rho_n : \text{IPPM}_n^- \longleftrightarrow \mathbf{I}_n^-(-, -, =) \quad \text{such that} \quad v(Q) = \text{dist}(\rho_n(Q)).$$

We give both algorithms and prove their inverseness. The common object is a collection of nonempty sets arranged around an oriented circle. There is no root to choose; a largest label records the terminal condition. After the original problem is solved, the same construction gives the full classes and retains the complete column-parity information.

2 Matrices drawn as decreasing blocks on a circle

A cyclically ordered set partition is an ordered list of nonempty disjoint sets, considered up to rotation but not reflection. Write $\mathcal{C}_{h,q}$ for such partitions of $[h]$ into q blocks. When a linear representative is needed, we always start at the block containing 1.

First observe that an improper column, in its local labels, has the form

$$(1, 2), (3, 4), \dots,$$

with each displayed pair in a single cell; an odd column has one final unpaired label. Indeed, a row change after local label $2r - 1$ is exactly a proper ascent or descent. This proves the equivalence with the original definition, rather than adding a new restriction.

Collapse each pair to one label and keep the possible final singleton. Relabel consecutively by columns. The resulting partition matrix P has $k = v(Q)$ labels; write S for its set of originally odd columns and $D = \dim P$. No cell disappears. If a collapsed column has m labels, it had $2m - 1$ labels when its index is in S , and $2m$ otherwise. Hence

$$n = 2k - |S|. \quad (1)$$

Conversely, double every label of P in its cell, then omit the second copy of the last label in each column of S . This is the inverse collapse. In particular, Q is minus precisely when $D \in S$: every cell in its last column consists of pairs, apart from the possible final singleton.

Here is how to draw P and S on a circle. Let c_i be the smallest label in column i of P . Process the labels $a = 1, \dots, k$ in order. If a lies in row i , insert a immediately before $c_i - 1$ in the current word; if $c_i - 1 = 0$, append a instead. The insertion is legal: upper triangularity implies $c_i \leq a$, so a positive $c_i - 1$ is already present. Call the final permutation σ .

The positive numbers $c_i - 1$ are exactly the descent-bottom values of σ . To see this, the first smaller letter to the right of a in the final word is its insertion anchor, or there is none if it was appended. Subsequent insertions are larger and cannot change this fact. If a positive anchor b is used, the last letter inserted before b remains immediately before it and gives a descent. Conversely an adjacent descent $a > b$ has first smaller right-hand letter b , so b was an anchor. Every row is nonempty, so every $c_i - 1$ occurs as an anchor. For an arbitrary permutation, the converse reconstruction also follows by restricting its word successively to $[1], [2], \dots, [k]$: the new letter is immediately before its first smaller right-hand letter, or is last if there is none.

Place the new largest letter $k + 1$ before σ and regard the word

$$\omega = (k + 1) \sigma$$

as circular. There is now one descent edge for each column. For $j < D$, column j corresponds to the descent whose bottom is its largest label $c_{j+1} - 1$. Column D corresponds to the edge leaving $k + 1$. Keep precisely the edges corresponding to S , and cut all other edges. Every kept edge descends, so each resulting block is decreasing. Forget the order within blocks, retaining their circular order. Denote the result by $M(Q)$. There is at least one cut: the edge entering $k + 1$ ascends. By (1), the number of blocks is

$$q = k + 1 - |S| = n - k + 1. \quad (2)$$

The block containing $k + 1$ is nonsingleton exactly when column D was odd.

Reading the matrix back

Given $C \in \mathcal{C}_{k+1,q}$, write each block decreasingly and start the circle at $k + 1$. Delete $k + 1$ from the resulting word, obtaining σ . For each letter a , let f_a be the first smaller letter to its right, or zero if there is none. Sort the distinct values as

$$0 = y_1 < y_2 < \dots < y_D, \quad y_{D+1} = k.$$

The collapsed matrix is explicitly

$$P_{ij} = \{a : f_a = y_i, y_j < a \leq y_{j+1}\}. \quad (3)$$

Each row occurs because its value is in the alphabet, and each column is a nonempty interval. Since $f_a < a$, a label in column j must have row index at most j , proving upper triangularity. This also shows that the column starts are $y_j + 1$, so insertion recovers σ .

Recover the odd columns by asking whether their associated descent edges lie inside blocks of C . For columns $j < D$ the bottom is y_{j+1} ; for column D the edge leaves $k + 1$. Expand these columns oddly and the others evenly by the inverse collapse already described.

These operations are inverse in both directions: first-smaller-right anchors invert insertion, block interiors recover exactly the kept edges, and the column expansions invert collapse. We have proved the following useful encoding, with no unconstructed bijection as an input.

Lemma 1. *For $k \geq 1$ and $k \leq n \leq 2k$, the map M is a bijection*

$$\{Q \in \text{IPPM}_n : v(Q) = k\} \longleftrightarrow \mathcal{C}_{k+1, n-k+1}.$$

Its restriction to minus matrices is exactly the subset whose largest element belongs to a nonsingleton block.

3 Inversion sequences on the same circle

We next give one recursion, removing the entire final run at each step. The empty sequence corresponds to the one-block partition $(\{1\})$. Suppose a nonempty restricted sequence has the form

$$e = e' a^\varepsilon, \quad \varepsilon \in \{1, 2\},$$

where a does not occur in e' . Set $L = |e'|$ and $k = \text{dist}(e)$. List the available new values in increasing order:

$$U = \{0, 1, \dots, L\} \setminus \text{Val}(e').$$

They are precisely the values allowed at the first position of this run. Recursively encode e' as a cycle C' , and start its blocks at the one containing 1. There are $L - k + 2$ blocks, the same as $|U|$. If a is the j th member of U , select the j th block. Now:

- if $\varepsilon = 1$, put $k + 1$ into the selected block;
- if $\varepsilon = 2$, put the singleton block $\{k + 1\}$ immediately before the selected block.

Keep circular order. Call the resulting cycle $B(e)$.

This is a single full-class rule. In particular, the double-run case does not call a separate minus bijection or first remove only the duplicate. Only the natural order of unused values within the current object is used.

The inverse recursion

The cycle $(\{1\})$ decodes as empty. Otherwise let $k + 1$ be the largest letter. If its block is nonsingleton, remove $k + 1$, remember the remaining block as the selected one, and set $\varepsilon = 1$. If its block is a singleton, remove that block, remember the next block in circular order, and set $\varepsilon = 2$. At least one block remains in this second case.

Decode the smaller cycle to e' . Start that cycle at its block containing 1 and find the index j of the selected block. Append ε copies of the j th member of $\{0, \dots, |e'|\} \setminus \text{Val}(e')$. This finishes the inverse algorithm.

Lemma 2. *For $k \geq 0$ and $1 \leq q \leq k + 1$, the map B is a bijection between restricted inversion sequences with k distinct values and length $k + q - 1$, and $\mathcal{C}_{k+1, q}$. When $k = 0$ this means only the empty sequence and $(\{1\})$. A nonempty sequence is minus exactly when the largest letter of its cycle lies in a nonsingleton block.*

Proof. Induct on k . For a prefix with $k - 1$ values and length L , induction gives $L - k + 2$ blocks. All its values lie in $\{0, \dots, L\}$, so the unused list also has $L - k + 2$ members. The selection is therefore defined. A singleton run increases the number of labels by one and leaves the block number unchanged. A double run increases both by one. In either case the output parameters are $k + 1$ labels and $|e| - k + 1$ blocks.

Conversely, deletion of the largest letter is unambiguous. In the nonsingleton case the smaller cycle has q blocks; in the singleton case it has $q - 1$. Thus induction supplies a prefix of length $k + q - 2$ or $k + q - 3$, respectively. Adding the prescribed run gives length $k + q - 1$. The appended value is unused and less than $|e'| + 1$, so both the inversion bound and the restricted-repeat condition hold. The selected block and the rank in the same unused list are restored exactly. This proves both inverse compositions, and the block containing the largest letter records the final run length. $\square$

4 Solution of the original problem

Theorem 3. *For every $n \geq 1$, the explicitly constructed maps*

$$\rho_n = B^{-1} \circ M, \quad \rho_n^{-1} = M^{-1} \circ B$$

solve Chern–Fu’s original minus problem. They are mutually inverse and preserve $v = \text{dist}$.

Proof. Fix $k = v(Q)$. Lemma 1 sends a minus matrix to a cycle on $[k + 1]$ with $n - k + 1$ blocks and with the largest letter in a nonsingleton block. Lemma 2 decodes precisely those cycles as minus sequences of length $k + (n - k + 1) - 1 = n$ with k distinct values. Their proved inverses give both compositions. At $n = 1$ the full route is

$$[\{1\}] \mapsto (\{1, 2\}) \mapsto (0).$$

Thus the endpoint is included, and every step of both maps is specified. $\square$

For comparison with the released construction [2], remove $k + 1$ from its nonsingleton block and start the remaining cycle there. This gives the ordered partition used in that construction. On the matrix side these are the decreasing pieces of σ cut at every ascent and at the unselected descents. On the sequence side the start is precisely the block selected by the unused final value. Consequently our minus map is the released minus map expressed with one largest sentinel; it is not an assumption of that map’s correctness.

5 The full classes and the information retained

The largest-block condition can now simply be removed.

Theorem 4. *The same two algorithms give a bijection*

$$\text{IPPM}_n \longleftrightarrow \mathbf{I}_n(-, -, =)$$

for every $n \geq 1$, preserving $v = \text{dist}$. Odd last-column size corresponds to a final singleton run, and even last-column size to a final double run. It restricts to Theorem 3 without modification.

Proof. Both full classes with parameter k are bijective to all of $\mathcal{C}_{k+1, n-k+1}$ by the two lemmas. The largest letter is either nonsingleton or singleton, which gives exactly the two stated terminal states. No terminal object is deleted and then passed to an assumed minus bijection. $\square$

There is more information on the circle than its block number. Write its blocks decreasingly and count descents around the entire circular word; call this number $\text{cdes}(C)$. Start the word at its largest letter. Order the other descent edges by their bottom values, and place the edge leaving the largest letter last. Record a 1 for an edge inside a block and a 0 for an edge between blocks. This gives an ordered binary word, called the *descent-cut signature*.

Theorem 5. For $C = M(Q)$, its circular descent count is $\dim Q$, and its descent-cut signature is exactly

$$(N_1 \bmod 2, \dots, N_D \bmod 2).$$

Thus the bijection retains every column parity, in its original order, as well as dimension and terminal state. For fixed $k \geq 1$, dimension D , and any binary word η of length D , the number of matrices with $v(Q) = k$ and this parity signature is

$$\left\langle \begin{matrix} k \\ D-1 \end{matrix} \right\rangle, \quad (4)$$

where the bracket counts permutations of $[k]$ with $D-1$ descents. Their weight is necessarily $2k - |\eta|$, where $|\eta|$ is the number of ones in η .

Proof. The circular word has the descents of σ and one additional descent leaving its largest letter; its entering edge ascends. The anchor argument gives $D-1$ descents in σ . Their increasing bottom values are $c_2 - 1, \dots, c_D - 1$, so the prescribed order is exactly column order, followed by column D . A column was odd exactly when its edge was kept. This proves both recovery statements.

Conversely, take any permutation with $D-1$ descents and any prescribed η . Its first-smaller-right anchors give a unique collapsed matrix of dimension D , and η gives a unique expansion. Hence each such permutation contributes one matrix, proving (4). $\square$

On sequences these refinements are read from their explicit cycle $B(e)$. We do not assert that the resulting dimension statistic is one of the usual elementary inversion-sequence statistics. The common circular object is the natural setting for the refinement.

Corollary 6. Put $t = 2k - n$ and let ε be 1 in the minus class and 0 in the complementary class. The number of objects of weight n , semi-weight k , dimension D , and terminal state ε is

$$\left\langle \begin{matrix} k \\ D-1 \end{matrix} \right\rangle \binom{D-1}{t-\varepsilon}. \quad (5)$$

Without fixing the state it is $\left\langle \begin{matrix} k \\ D-1 \end{matrix} \right\rangle \binom{D}{t}$. Without fixing dimension the full number is

$$(n-k)! \left\{ \begin{matrix} k+1 \\ n-k+1 \end{matrix} \right\}, \quad (6)$$

where braces are Stirling numbers of the second kind. The minus number is $(n-k+1)! \left\{ \begin{matrix} k \\ n-k+1 \end{matrix} \right\}$.

Proof. For (5), choose the $t - \varepsilon$ odd columns among the first $D-1$ and apply Theorem 5. There are $(q-1)! \left\{ \begin{matrix} k+1 \\ q \end{matrix} \right\}$ cyclic partitions into q blocks. For the minus class, remove the largest letter from its nonsingleton block. What remains is a partition of $[k]$ with a circular order and a chosen starting block, hence an ordered partition into q blocks. Substitute $q = n - k + 1$. $\square$

All bounds are included. There are no objects unless $k \leq n \leq 2k$ and $1 \leq D \leq k$; in a specified dimension one also needs $0 \leq 2k - n \leq D$. For a specified terminal state, the sharper condition is $0 \leq 2k - n - \varepsilon \leq D - 1$. Outside these ranges the binomial coefficient is zero. At $n = k$ all columns must have odd size one, forcing the unique diagonal matrix; at $n = 2k$ all columns are even and there are $k!$ objects. The empty object may be adjoined on both sides, represented by $(\{1\})$, with dimension and semi-weight zero.

This reaches the full boundary of the binary-parity construction: every cyclic partition occurs, and every allowed ordered parity signature occurs. No claim is made about runs of length three or more, which violate the original avoidance condition and would require a different common class.

6 Example

Consider

$$Q = \begin{pmatrix} \{1\} & \{2, 3, 6\} \\ & \{4, 5\} \end{pmatrix}.$$

Collapse to $P_{11} = \{1\}$, $P_{12} = \{2, 4\}$, $P_{22} = \{3\}$, with both columns odd. The insertion word is 3124. On the circle 53124, keep $5 \rightarrow 3$ and $3 \rightarrow 1$, and cut the other edges. Thus

$$M(Q) = (\{1, 3, 5\}, \{2\}, \{4\}).$$

The inverse sequence recursion successively removes 5, then the singleton 4, then 3, then the singleton 2. It yields

$$(0, 0, 1, 2, 2, 3).$$

The circular word has two descents, both kept, recovering dimension two and the signature $(1, 1)$. This example illustrates the full algorithm, not an invocation of a precomputed minus map.

7 Attribution

The original problem, semi-weight, and endpoint classes are due to Chern–Fu [1]. Their Lemma 5.6 already shows that the minus and full existence problems are equivalent; that reduction is not presented as a new result here. The pair-collapse, insertion, and ordered/cyclic partition ingredients come from the released proof [2], and the collapsed-matrix anchors are the Claesson–Dukes–Kubitzke correspondence recalled in Chern–Fu, Theorem 1.2. We have given their needed proofs and reorganized the construction as two directly invertible full-cycle maps. The single final-run recursion replaces the two mutually recursive encoders. The parity signature and circular descents describe exactly what the construction retains. No priority claim is made for these refinements; the Stirling enumeration was already known in the source literature.

References

- [1] S. Chern and S. Fu, *Signed counting of partition matrices*, Question 5.7, Definitions 1.1, 2.1, 3.1, and Lemma 5.6. [arXiv:2508.21318v2](https://arxiv.org/abs/2508.21318v2).
- [2] EULER public proof, [An explicit statistic-preserving bijection](#), especially its parity collapse, insertion permutation, and cyclic-partition encoders. The present note is a self-contained reorganization and refinement of those ingredients.